

Social Media Data Analysis

Lecture 1: Introduction + Potentials and Pitfalls of Social Media Data Analysis

Indira Sen, FSS 2026

A little bit about the instructors



Indira Sen

Junior Faculty, background in
Computer Science, research on
societal impacts of Large
Language Models (LLMs),
Human-LLM interaction

A little bit about the instructors



Georg Ahnert

PhD student, background in
Social Data Science,
research on LLMs & public
opinion surveys



Maximilian Kreutner

PhD student, background
in Computer Science and
Artificial Intelligence
Engineering, research on
LLMs & social simulation

Chair of Data Science for the Economic and Social Sciences (DESS)



A little bit about you



About this course

Materials: on ILIAS or Github*

<https://tinyurl.com/uma-smda>

Why do we study society and social phenomena?

- ❖ To learn about society
 - How do sexist attitudes propagate?
 - Do disease outbreaks also affect mental health?
- ❖ To design policies and interventions for addressing societal problems
 - Hate speech
 - Polarization

How do we study society and social phenomena?

- ❖ To learn about society
 - How do sexist attitudes propagate?
 - Do disease outbreaks also affect mental health?



Questionnaire
Design



Solicitation of
Responses

Surveys

- ❖ To design policies and interventions for addressing societal problems
 - Hate speech
 - Polarization



Manual
analysis

How do we study society and social phenomena?

- ❖ To learn about society
 - How do sexist attitudes propagate?
 - Do disease outbreaks also affect mental health?



Questionnaire
Design



Solicitation of
Responses

Surveys

- Declining response rates
- Reporting biases
- ...

- ❖ To design policies and interventions for addressing societal problems
 - Hate speech
 - Polarization



Manual
analysis

- Emotional and psychological trauma
- Intractable at scale
- ...

Are there other ways of measuring
social phenomena?

Are there other ways of measuring
social phenomena?

**Enter: web and social media data
+ computational methods**



Enter: web and social media data
+ computational methods

The image shows a screenshot of a tweet from Jon Snow (@kingofthenorth) dated June 1, 2021. The tweet text is: "*yawn* Im sorry but women cannot drive, call me sexist or whatever but it is true." The phrase "women cannot drive," is highlighted with a red box. A blue arrow points from a box labeled "Input: Text to be labeled" to this red box. A red arrow points from the red box to a box labeled "Output: Sexist; stereotypes".

Jon Snow
@kingofthenorth

yawn Im sorry but women cannot drive, call me sexist or whatever but it is true.

12:00 PM · Jun 1, 2021

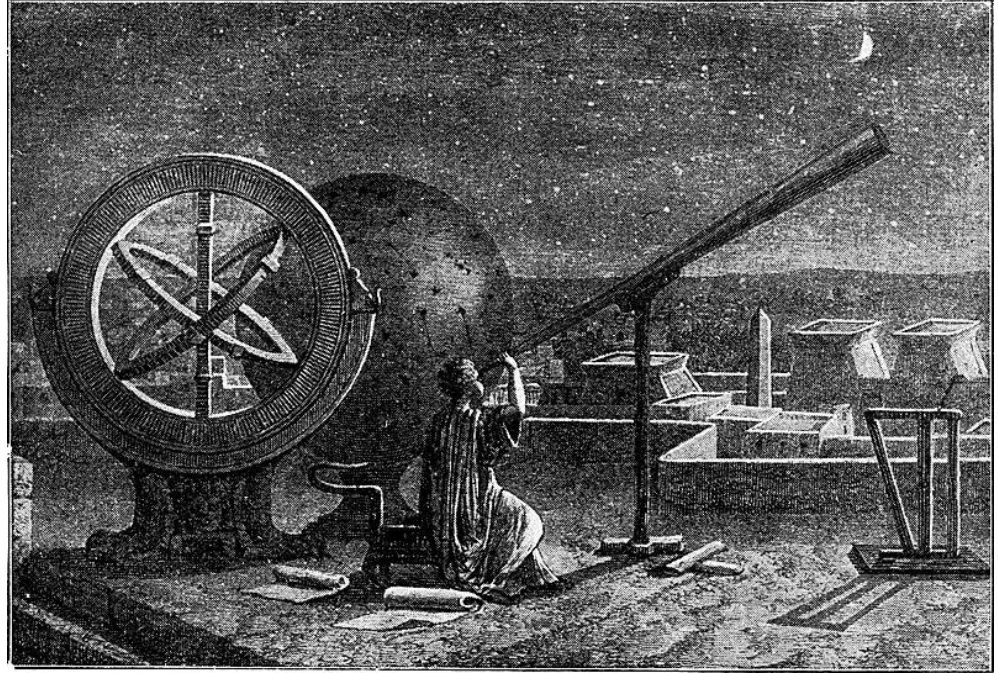
64 Retweets 32 Quote Tweets 422 Likes

Input: Text to be labeled

Output: Sexist; stereotypes

Enter: web and social media data
+ computational methods

A Computational Turn: A Measurement Revolution?



Computational Social Science

David Lazer,¹ Alex Pentland,² Lada Adamic,³ Sinan Aral,^{2,4} Albert-László Barabási,⁵ Devon Brewer,⁶ Nicholas Christakis,¹ Noshir Contractor,⁷ James Fowler,⁸ Myron Gutmann,³ Tony Jebara,⁹ Gary King,¹ Michael Macy,¹⁰ Deb Roy,² Marshall Van Alstyne^{2,11}

A field is emerging that leverages the capacity to collect and analyze data at a scale that may reveal patterns of individual and group behaviors.

Download

We live life in the network. We check our e-mails regularly, make mobile phone calls from almost any location, swipe transit cards to use public transportation, and make purchases with credit cards. Our movements in public places may be captured by video cameras, and our medical records stored as digital files. We may post blog entries accessible to anyone, or maintain friendships through online social networks. Each of these transactions leaves digital traces that can be compiled into comprehensive pictures of both individual and group behavior, with the potential to transform our understanding of our lives, organizations, and societies.

The capacity to collect and analyze massive amounts of data has transformed such fields as biology and physics. But the emergence of a data-driven “computational social science” has been much slower. Leading journals in economics, sociology, and political science show little evidence of this field. But computational social science is occurring—in Internet companies such as Google and Yahoo, and in govern-

NATURE | Vol 445 | 1 February 2007

A twenty-first century science

If handled appropriately, data about Internet-based communication and interactivity could revolutionize our understanding of collective human behaviour.

Duncan J. Watts

Few would deny that many of the major problems currently facing humanity are social and economic in nature. From the apparent wave of religious fundamentalism sweeping the Islamic world (and parts of the Western world), to collective economic security, global warming and the great epidemics of our times, powerful yet mysterious social forces come into play.

But few readers of *Nature* would consider social science to be the science of the twenty-first century. Although economics, sociology, political science and anthropology have produced a plethora of findings regarding human social behaviour, they

self-reports from participants, which suffer from cognitive biases, errors of perception and framing ambiguities.

The striking proliferation over the past decade of Internet-based communication and interactivity, however, is beginning to lift these constraints. For the first time, we can begin to observe the real-time interactions of millions of people at a resolution that is sensitive to effects at the level of the individual. Meanwhile, ever-faster computers permit us to simulate large networks of social interactions. The result has been tremendous interest in social networks: thousands of papers and a growing number of books have been published less than a decade, leading some to

framework of collective social dynamics. People do not just interact: their interactions have consequences for the choices they, and others, make.

Studies that combine all these features are currently beyond the state of the art, but two of my group's recent projects indicate tentative progress. The first used the anonymized e-mail logs of a university community of around 40,000 people to track daily network evolution over a year as a function of existing network structure, shared activities (such as classes) and individual attributes. Dynamic data of this type may shed light on the relative roles of struc-



¹Harvard University, Cambridge, MA, USA. ²Massachusetts Institute of Technology, Cambridge, MA, USA. ³University of Michigan, Ann Arbor, MI, USA. ⁴New York University, New York, NY, USA. ⁵Northeastern University, Boston, MA, USA. ⁶Interdisciplinary Scientific Research, Seattle, WA, USA. ⁷Northwestern University, Evanston, IL, USA. ⁸University of California—San Diego, La Jolla, CA, USA. ⁹Columbia University, New York, NY, USA. ¹⁰Cornell University, Ithaca, NY, USA. ¹¹Boston University, Boston, MA, USA. E-mail: david_lazer@harvard.edu. Complete affiliations are listed in the supporting online material.

Data from the blogosphere. Shown is a link structure within a community of political blogs (from 2004), where red nodes indicate conservative blogs, and blue liberal. Orange links go from liberal to conservative, and purple ones from conservative to liberal. The size of each blog reflects the number of other blogs that link to it. [Reproduced from (8) with permission from the Association for Computing Machinery]

Reference:

[Computational social science](#)
[A Twenty-first Century Science](#)

Computational Social Science

David Lazer,¹ Alex Pentland,² Lada Adamic,³ Sinan Aral,^{2,4} Albert-László Barabási,⁵ Devon Brewer,⁶ Nicholas Christakis,¹ Noshir Contractor,⁷ James Fowler,⁸ Myron Gutmann,³ Tony Jebara,⁹ Gary King,¹ Michael Macy,¹⁰ Deb Roy,² Marshall Van Alstyne^{2,11}

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But few readers of *Nature* would consider social science to be the science of the twenty-first century. Although economics, sociology, political science and anthropology have produced a plethora of findings regarding human social behaviour, they

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Digital Traces + computational methods are the ***Telescope*** for the Social Sciences

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Computational Social Science

SOCIAL SCIENCE

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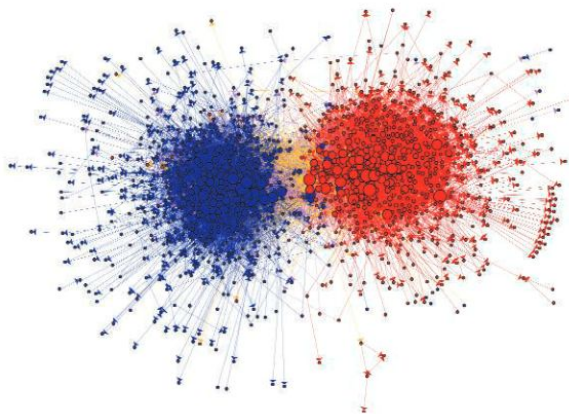
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ment agencies such as the U.S. National Security Agency. Computational social science could become the exclusive domain of private companies and government agencies. Alternatively, there might emerge a privileged set of academic researchers presiding over private data from which they produce papers that cannot be

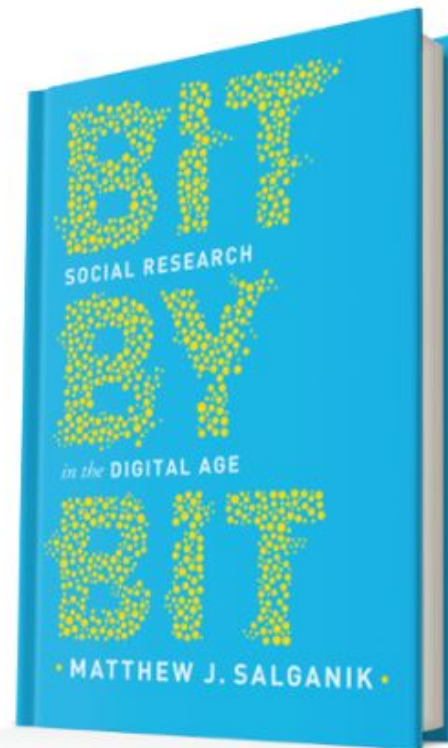
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critiqued or replicated. Neither scenario will serve the long-term public interest of accumulating, verifying, and disseminating knowledge.

What value might a computational social science—based in an open academic environment—offer society, by enhancing understanding of individuals and collectives? What are the



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Salganik, Matthew J. *Bit by bit: Social research in the digital age*. Princeton University Press, 2019.

What is Social Data Science?

The aim of Social Data Science is: **Understanding** of **Social Phenomena** using **Data Science**.

- **Understanding**: Not just predicting, we want to be able to generalize and combine knowledge, and even to motivate interventions or policies.
- **Social Phenomena**: ABCs of humans and society — Attitudes, Behaviors, and Characteristics
- **Data Science**: Using statistical methods, machine learning, **Large Language Models!**

What is social media?

2021 *This Is What Happens In An Internet Minute*



What does social media have to do with (social) science?

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Social Media can advance Social Science in the following ways:

1. Use Social Media to learn *about* society

Web Search Queries Can Predict Stock Market Volumes

Ilaria Bordino, Stefano Battiston, Guido Caldarelli, Matthieu Cristelli, Antti Ukkonen, Ingmar Weber

Published: July 19, 2012 • <https://doi.org/10.1371/journal.pone.0040014>

Article	Authors	Metrics	Comments	Media Coverage
				
Abstract Introduction Results Discussion Materials and Methods Supporting Information Acknowledgments Author Contributions References Reader Comments Figures	Abstract We live in a computerized and networked society where many of our actions leave a digital trace and affect other people's actions. This has led to the emergence of a new data-driven research field: mathematical methods of computer science, statistical physics and sociometry provide insights on a wide range of disciplines ranging from social science to human mobility. A recent important discovery is that search engine traffic (i.e., the number of requests submitted by users to search engines on the www) can be used to track and, in some cases, to anticipate the dynamics of social phenomena. Successful examples include unemployment levels, car and home sales, and epidemics spreading. Few recent works applied this approach to stock prices and market sentiment. However, it remains unclear if trends in financial markets can be anticipated by the collective wisdom of on-line users on the web. Here we show that daily trading volumes of stocks traded in NASDAQ-100 are correlated with daily volumes of queries related to the same stocks. In particular, query volumes anticipate in many cases peaks of trading by one day or more. Our analysis is carried out on a unique dataset of queries, submitted to an important web search engine, which enable us to investigate also the user behavior. We show that the query volume dynamics emerges from the collective but seemingly uncoordinated activity of many users. These findings contribute to the debate on the identification of early warnings of financial systemic risk, based on the activity of users of the www.			

Bordino, Ilaria, et al. "[Web search queries can predict stock market volumes](https://doi.org/10.1371/journal.pone.0040014)." *PloS one* 7.7 (2012): e40014.

What does social media have to do with (social) science?

Social Media can advance Social Science in the following ways:

1. Use Social Media to learn *about* society
2. Study Social Media's *impact on* society

How do social media feed algorithms affect attitudes and behavior in an election campaign?

Andrew M. Guess^{1*}, Neil Malhotra², Jennifer Pan³, Pablo Barberá⁴, Hunt Allcott⁵, Taylor Brown⁴, Adriana Crespo-Tenorio⁴, Drew Dimmery^{4,6}, Deen Freelon⁷, Matthew Gentzkow³, Sandra González-Bailón⁹, Edward Kennedy¹⁰, Young Mie Kim¹¹, David Lazer¹², Devra Moehler⁴, Brendan Nyhan¹³, Carlos Velasco Rivera⁴, Jaime Settle¹⁴, Daniel Robert Thomas⁴, Emily Thorson¹⁵, Rebekah Tromble¹⁶, Arjun Wilkins⁴, Magdalena Wojcieszak^{17,18}, Beixian Xiong⁴, Chad Kiewiet de Jonge⁴, Annie Franco⁴, Winter Mason⁴, Natalie Jomini Stroud¹⁹, Joshua A. Tucker²⁰

We investigated the effects of Facebook's and Instagram's feed algorithms during the 2020 US election. We assigned a sample of consenting users to reverse-chronologically-ordered feeds instead of the default algorithms. Moving users out of algorithmic feeds substantially decreased the time they spent on the platforms and their activity. The chronological feed also affected exposure to content: The amount of political and untrustworthy content they saw increased on both platforms, the amount of content classified as uncivil or containing slur words they saw decreased on Facebook, and the amount of content from moderate friends and sources with ideologically mixed audiences they saw increased on Facebook. Despite these substantial changes in users' on-platform experience, the chronological feed did not significantly alter levels of issue polarization, affective polarization, political knowledge, or other key attitudes during the 3-month study period.

Guess, Andrew M., et al. "[How do social media feed algorithms affect attitudes and behavior in an election campaign?](#)" *Science* 381.6656 (2023): 398-404.

Is this only useful for scientific research?

Many industries do “**social media listening**”: identifying and assessing what is being said about a company, individual, product or brand on the internet.

Helps develop marketing campaigns

SCREEN	% mentions	Net sentiment	Social listening score
 Apple	0.92%	 39%	 36
SAMSUNG	5.5%	 70%	 100

Social Media Data *complements* Traditional Data Sources

Article

The Price of Power: Power Seeking and Backlash Against Female Politicians

Tyler G. Okimoto¹ and Victoria L. Brescoll¹

Personality and Social
Psychology Bulletin
36(7) 923–936
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DOI: 10.1177/0146167210371949
http://pspb.sagepub.com
SAGE

Abstract

Two experimental studies examined the effect of power-seeking intentions on backlash toward women in political office. It was hypothesized that a female politician's career progress may be hindered by the belief that she seeks power, as this desire may violate prescribed communal expectations for women and thereby elicit interpersonal penalties. Results suggested that voting preferences for female candidates were negatively influenced by her power-seeking intentions (actual or perceived) but that preferences for male candidates were unaffected by power-seeking intentions. These differential reactions were partly explained by the perceived lack of communality implied by women's power-seeking intentions, resulting in lower perceived competence and feelings of moral outrage. The presence of moral-emotional reactions suggests that backlash arises from the violation of communal prescriptions rather than normative deviations more generally. These findings illuminate one potential source of gender bias in politics.

Keywords

gender stereotypes, backlash, power, politics, intention, moral outrage

Okimoto et al., [The Price of Power: Power Seeking and Backlash Against Female Politicians](#)

- Uses a survey experiment where people are shown fictional politicians with different traits and asked to choose whom they prefer

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- Uses a survey experiment where people are shown fictional politicians with different traits and asked to choose whom they prefer

Full & Short Papers

WebSci '19, June 30–July 3, 2019, Boston, MA, USA

As the Tweet, so the Reply? Gender Bias in Digital Communication with Politicians

Armin Mertens
Cologne Center for Comparative Politics
Cologne, Germany
mertens@wiso.uni-koeln.de

Ayjeran Rozyjumayeva
Faculty of Management, Economics and Social Sciences
Cologne, Germany
rozyjumayeva@wiso.uni-koeln.de

Franziska Pradel
Cologne Center for Comparative Politics
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Jens Wäckerle
Cologne Center for Comparative Politics
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ABSTRACT

This study investigates gender bias in political interactions on digital platforms by considering how politicians present themselves on Twitter and how they are approached by others. Incorporating social identity theory, we use dictionary analyses to detect biases in individual tweets connected to the German federal elections in 2017. Besides sentiment analysis, we introduce a new measure of personal- vs. job-related content in text data, that is validated with structural topic models. Our results indicate that politicians' communication on Twitter is driven by party identity rather than gender. However, we find systematic gender differences in tweets

1 INTRODUCTION

"Why do you even care for the #BER [Berlin airport] opening? Because you have to choose your clothes for the opening ceremony???"

This tweet directed at a female German politician resembles an often observed and reported phenomenon in our digital society: digital communication is driven by gender stereotypes rather than job-related content. This, in turn, can lead to bias and discrimination towards female professionals. Building on insights from research on social identities [60, 61] as well as gender biases and gender roles in

Mertens et al., [As the Tweet, so the Reply? Gender Bias in Digital Communication with Politicians](#)

- Uses tweets mentioning real-life politicians and computationally assesses stereotypes in those tweets

Here are some questions we will explore in this course

- What are the components of a social media platform and how do they work?
- How do we collect and store data from these platforms?
- How do we **computationally** analyze this data?
- How do we validate the results of our social media analysis?
- How do we collect and analyze social media data in an ethical manner?

You should take this course because:

- You can scroll Tiktok for 1.5 hours (more)

You should take this course because:

- ~~You can scroll Tiktok for 1.5 hours (more)~~ You will learn how web and social media data from a variety of platforms, e.g., Youtube, Reddit, Wikipedia, Tiktok, can be used for social and economic use cases
- You will learn about Computational Social Science:
 - Main methods used in CSS
 - Applications of social media to CSS research questions and applications
 - CSS approaches to studying social media
- You will get experience with a hands-on CSS project that uses social media data that can help you for your Master Thesis

You should take this course, if you have:

- Strong knowledge of and experience in programming, ideally Python
 - All code provided from instructors will be in Python
 - You are expected to do the course project in Python
- An interest in scientific research
 - Ideally, you have read a scientific research article before
 - If not, please check: S.Keshav's "[How to Read a Paper](#)"

This course is based on recent, state-of-the-art research in Computational Social Science

You are expected to go over the readings assigned for each lecture and participate in discussions

Course Logistics

Lecture Topics/Weeks

1. Introduction + Why Study Social Media?

2. Data Collection 1: Web Scraping

3. Data Collection 2: APIs

4. Platform Affordances

5. Sampling Social Media Data + Project Discussions

6. Data Cleaning

7. Exploratory Data Analysis

8. Textual Analysis of Social Media Data I

9. Project Pitches

10. Textual Analysis of Social Media Data II

11. Network Analysis of Social Media Data

12. Ethics, Reproducibility and Documentation

13. Midway Project Presentations

14. Summary, Outlook, and Project Discussions

We have weekly lectures and exercises

- Lectures
 - Tuesday 1:45-3:15 PM
 - Starting from week 4: 10-20 minutes of group discussion at the end
- Exercises
 - Recap and **hands-on application** of what we saw in the lecture
 - Weekly exercise questions (**not graded**)
 - Solutions are presented in the exercise or the week after
 - Project Consultations
 - In the last two exercise sessions, you can work on your projects in the exercise slots
 - Wednesday 1:45-3:15 PM OR Thursday 1:45-3:15 PM

Grading

- Group Project (80%)
- Class Participation (20%)
 - Every lecture session will have 1-2 items announced ahead: mainly papers but could also be blog posts, videos, etc
 - You are expected to read or watch these before the lecture
 - 10-15 mins before the end of every lecture, we will have a short discussion
 - What you learned
 - What you agree or disagree with
 - How you can improve what you read
 -



Group Project

- **Equivalent to a scientific short paper: 4-5 pages, double column**
- You have to work in teams of 3-6; team size suggestion will be finalized based on how many people register for the course
- If you cannot find project partners: use the forum on ILIAS
 - Worst case, you'll be merged with another group
- We have some suggestions for projects ideas, but you are encouraged to come up with your own and be creative!

More about the Projects: Timeline and Dates

- Start forming groups: week 2
- Project pitches: 21.04.2026 [5%]
- Midway Presentation: 19.05.2026 [15%]
- Pre-final report submission: 22.06.2026
[optional and ungraded]: a chance to get
any critical feedback
- Final reports due: 07.07.2026 [60%]

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You cannot drop the course after the project pitches, so be sure to deregister earlier if you don't want to participate in the course

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You also cannot change groups after the project pitches

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More about the Projects: Presentations and Report

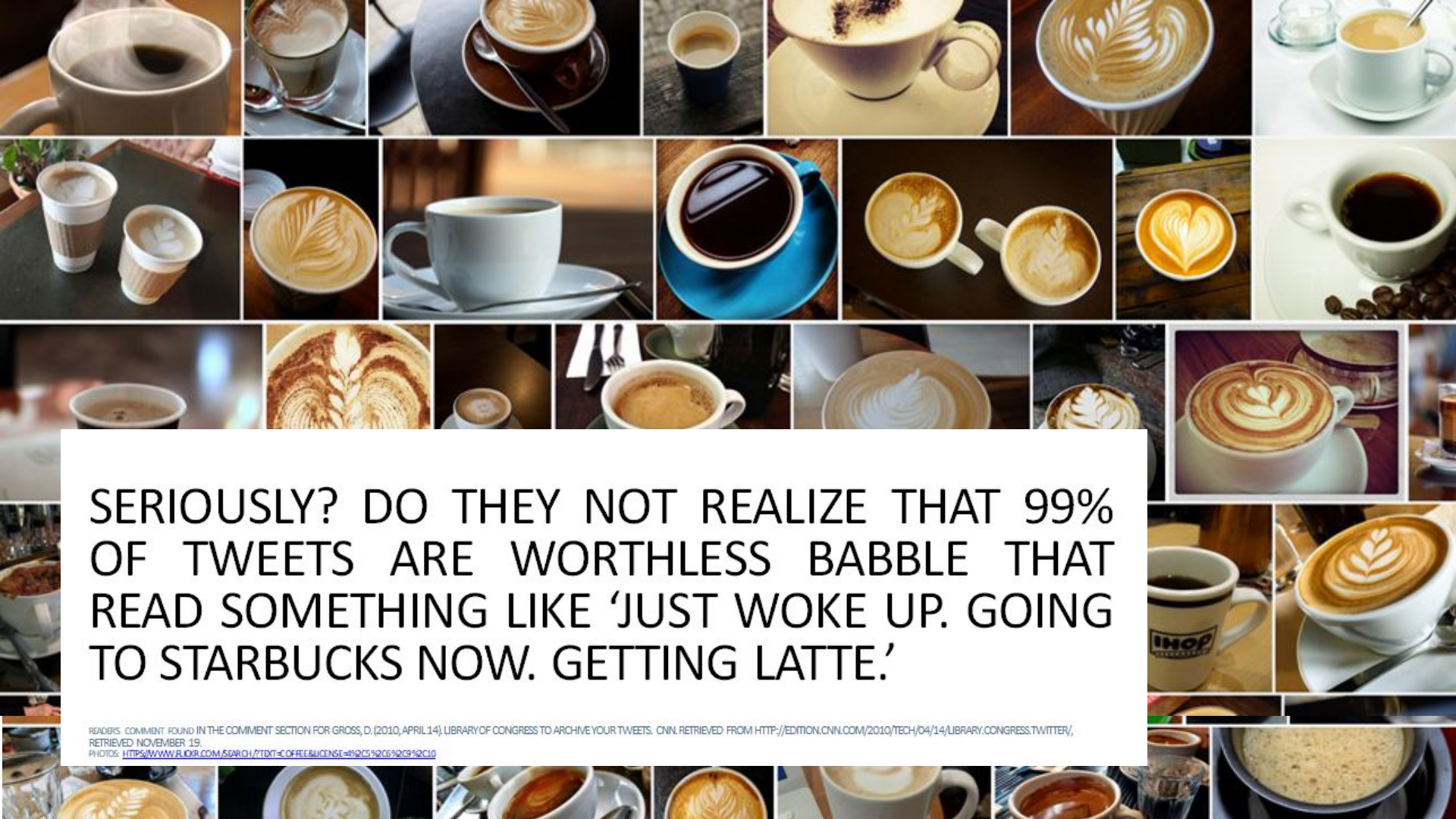
- Project pitches [5%] → 5 mins per teams
- Midway Presentation [15%] → 15 mins per team
- Final reports [60%]
 - All reports should include: introduction, related work, experiments, results, conclusion
 - Team member contributions: all members of a group need not get the same grades
 - Ideally: also include your code

Questions?

More details on grading is available on ILIAS/github page

Let's see what we
can learn from social
media data





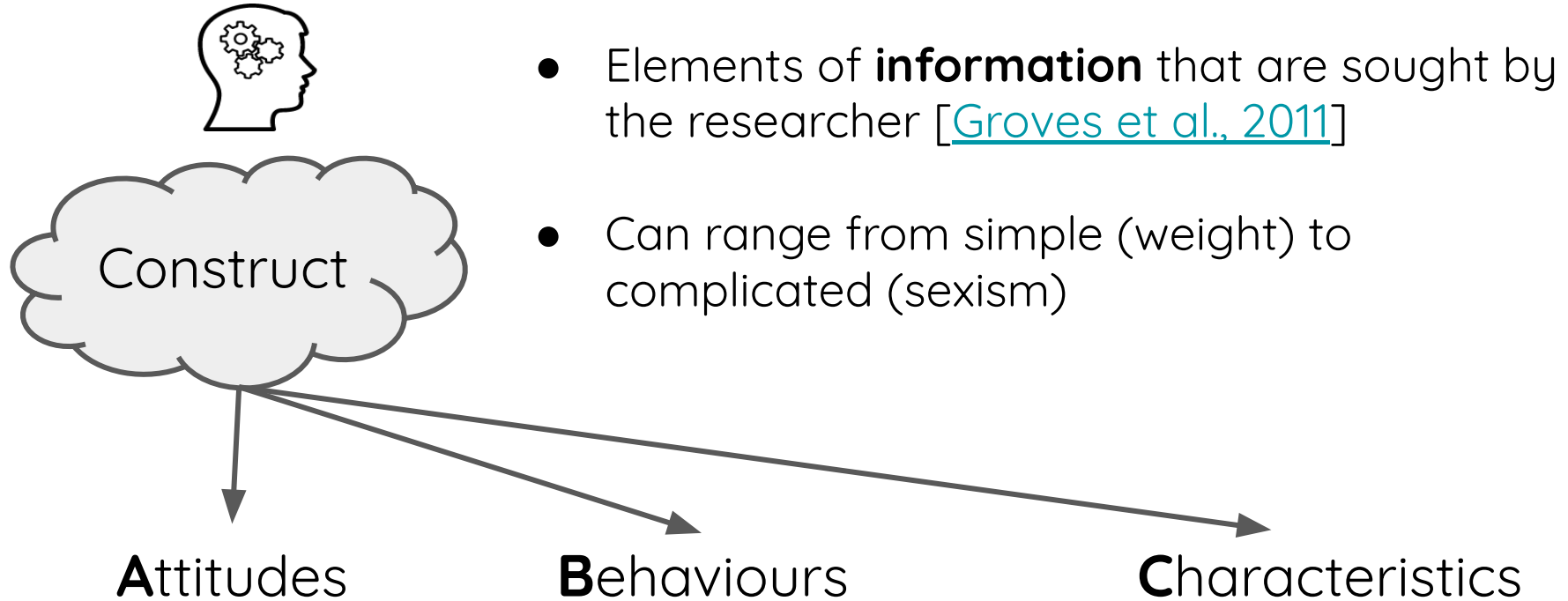
SERIOUSLY? DO THEY NOT REALIZE THAT 99% OF TWEETS ARE WORTHLESS BABBLE THAT READ SOMETHING LIKE 'JUST WOKE UP. GOING TO STARBUCKS NOW. GETTING LATTE.'

READER'S COMMENT FOUND IN THE COMMENT SECTION FOR GROSS, D. (2010, APRIL 14). LIBRARY OF CONGRESS TO ARCHIVE YOUR TWEETS. CNN. RETRIEVED FROM [HTTP://EDITION.CNN.COM/2010/TECH/04/14/LIBRARY.CONGRESS.TWITTER/](http://edition.cnn.com/2010/TECH/04/14/library.congress.twitter/), RETRIEVED NOVEMBER 19.
PHOTOS: [HTTP://WWW.FLICKR.COM/SEARCH/?TEXT=COFFEE&LICENSE=192559265926392610](http://www.flickr.com/search/?text=coffee&license=192559265926392610)

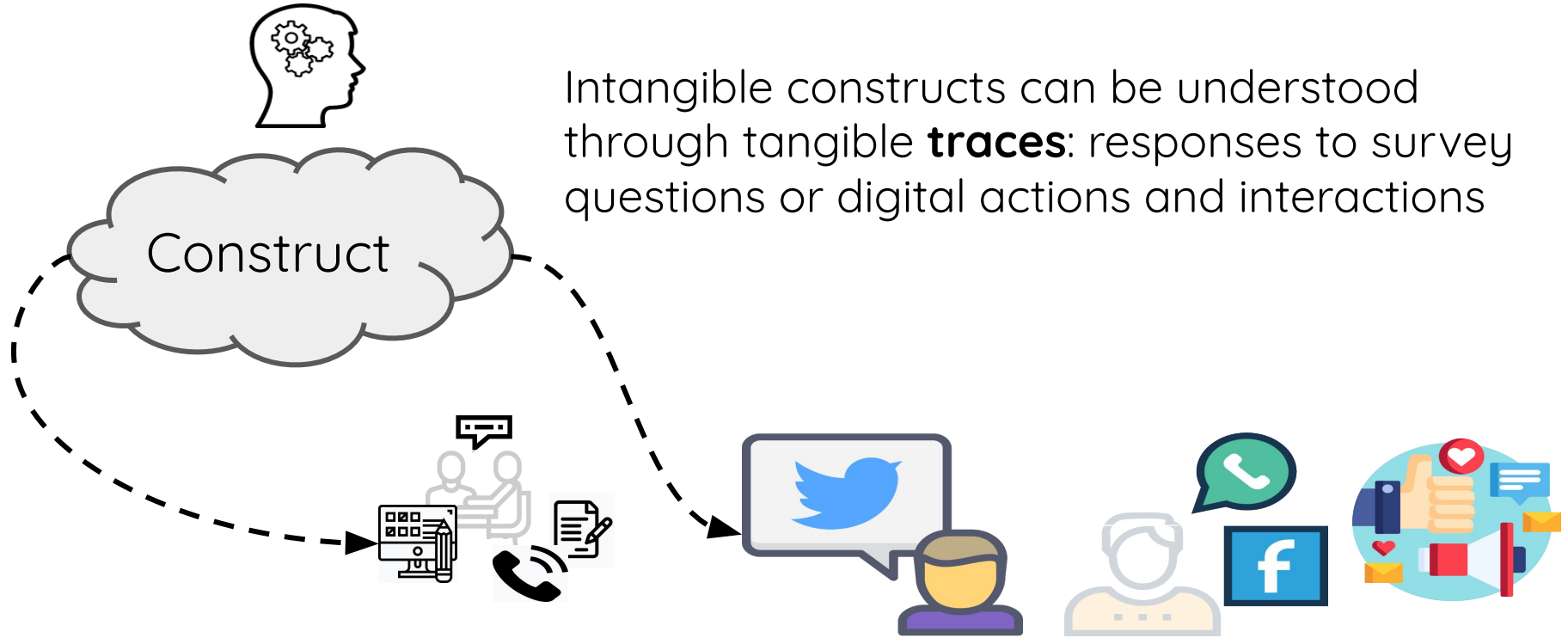
Understanding Social Phenomena



Understanding Social Phenomena

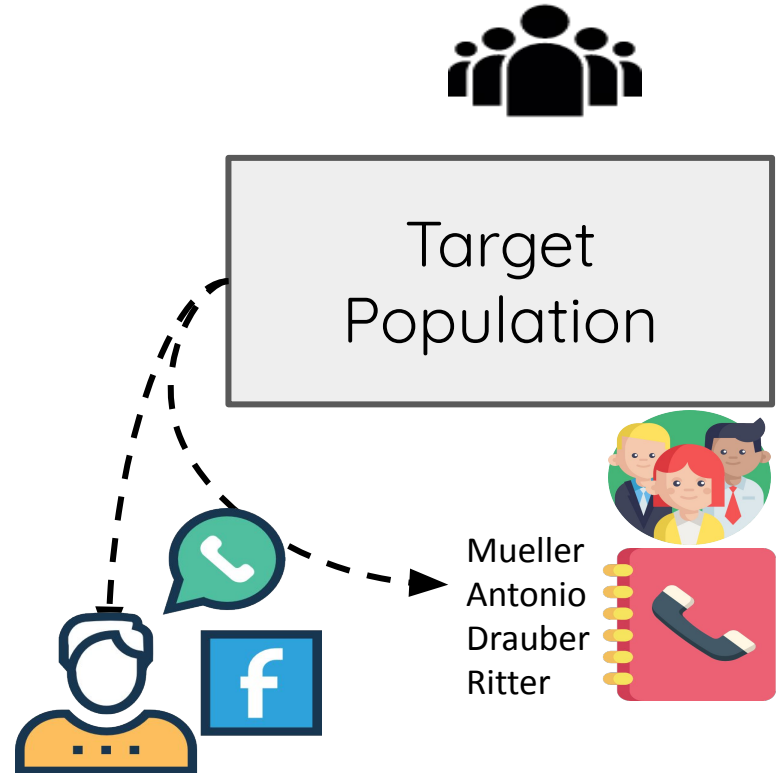


Understanding Social Phenomena



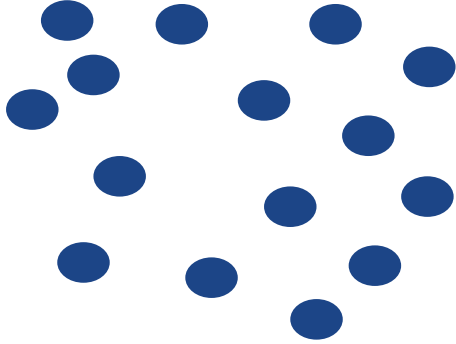
Understanding Social Phenomena

- **Members** of the target population (TP) are usually represented as elements in a record.
- This can be offline records (census, phonebook), online platforms (profiles, IPs), or other digital forms (bank accounts, fitness trackers)

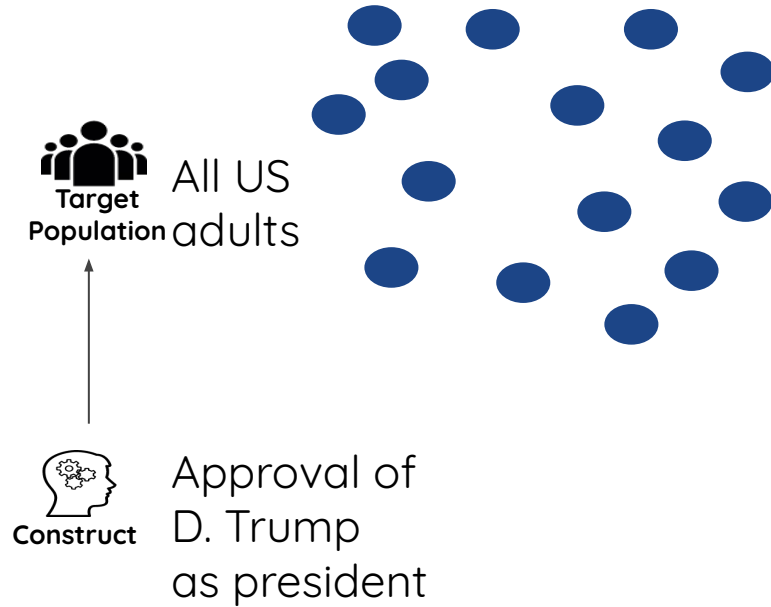


A typical research pipeline with digital trace data

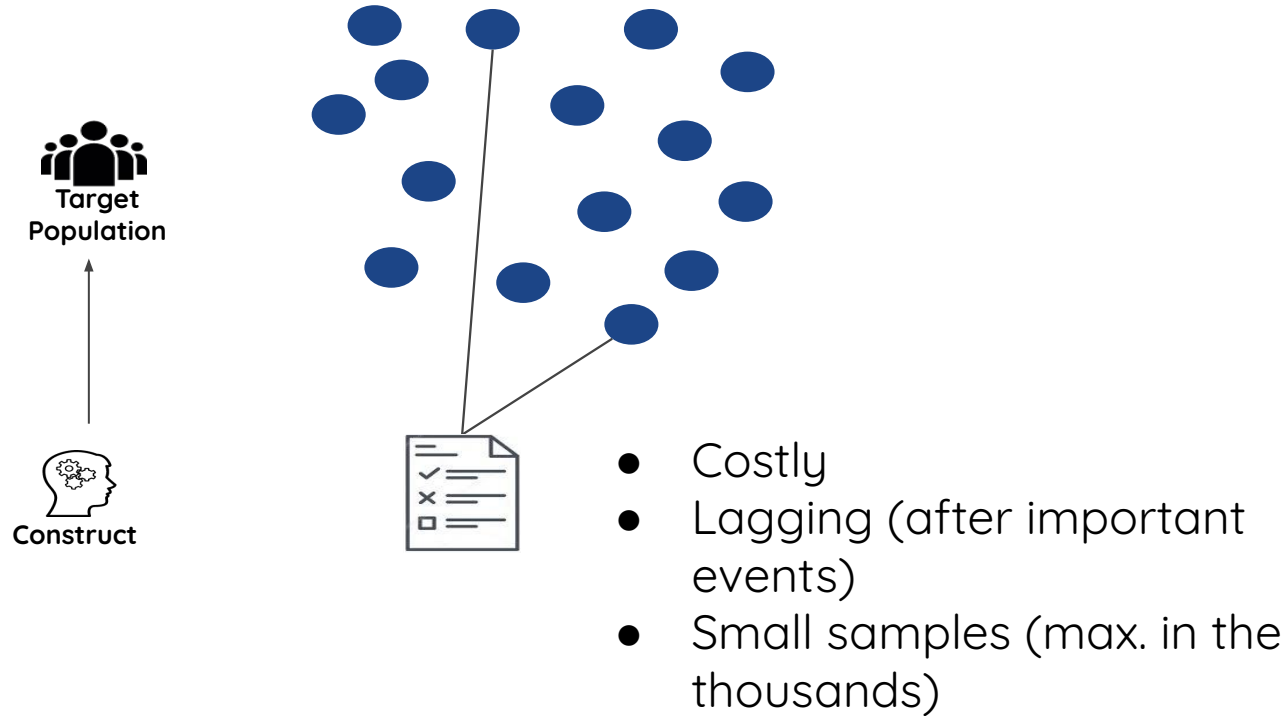
A typical research design with digital traces



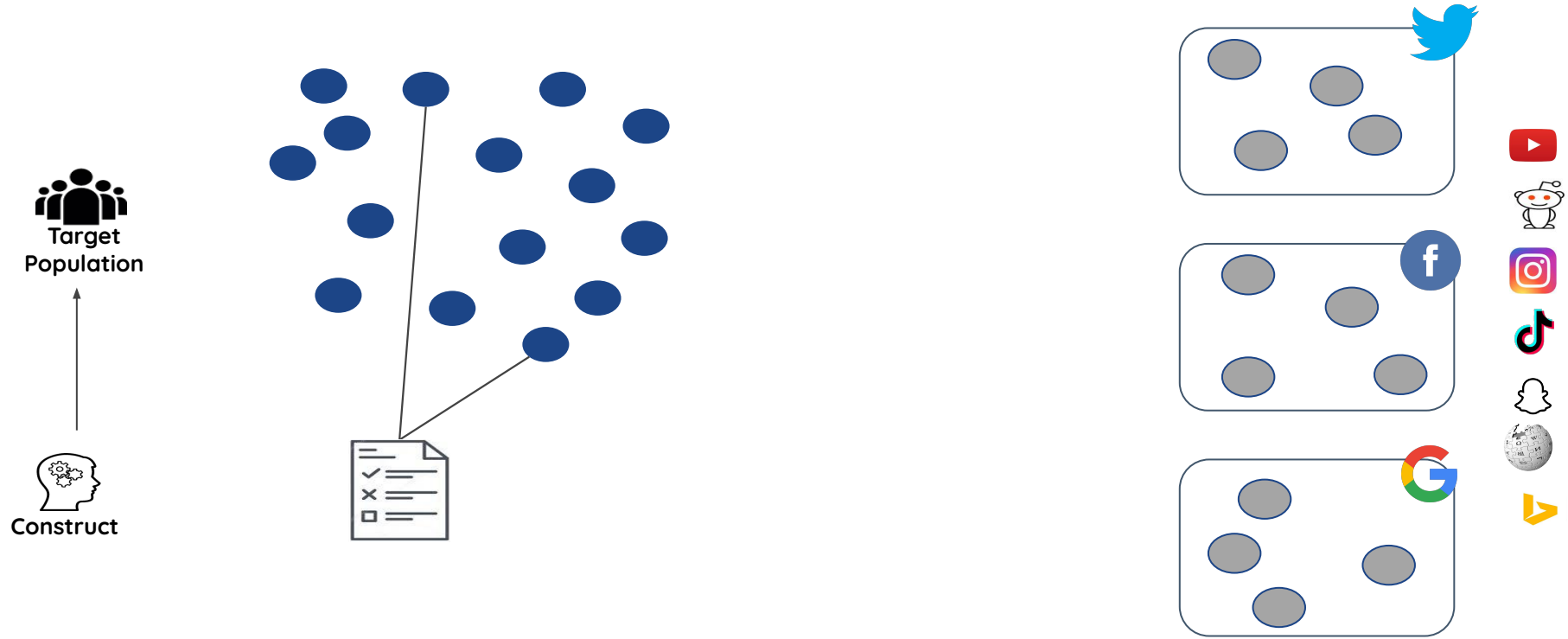
A typical research design with digital traces



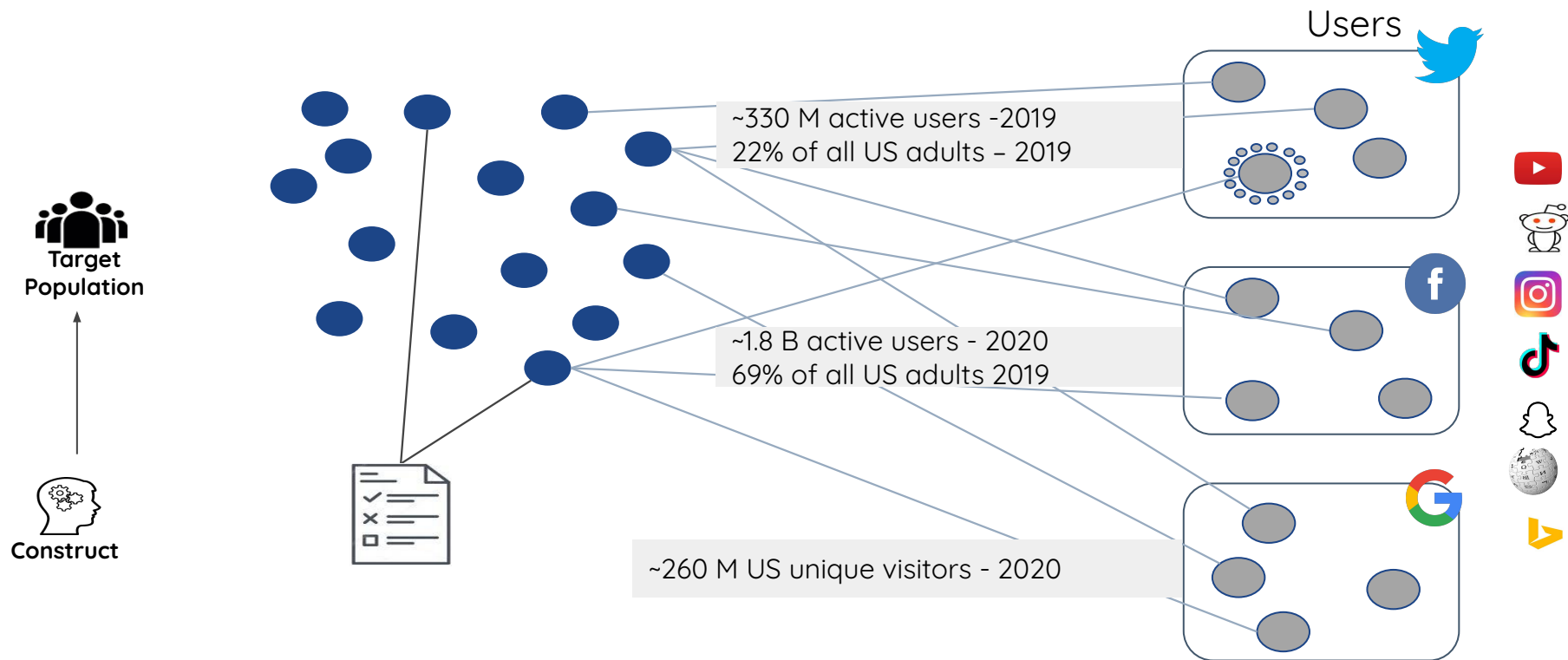
A typical research design with digital traces



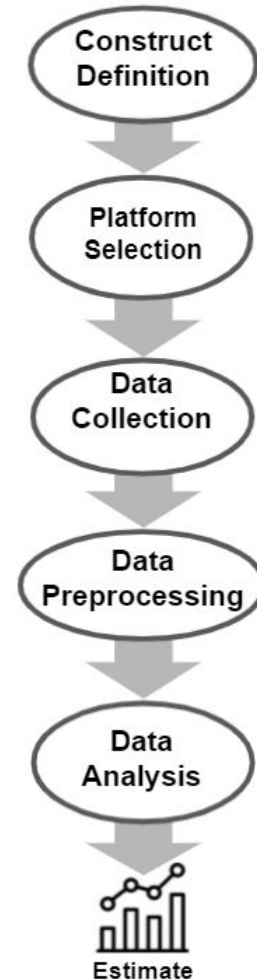
A typical research design with digital traces



A typical research design with digital traces



Prototypical Pipeline of Research with Social Media Data



Prototypical Pipeline: Data Collection

collect data potentially containing tangible signals regarding our construct

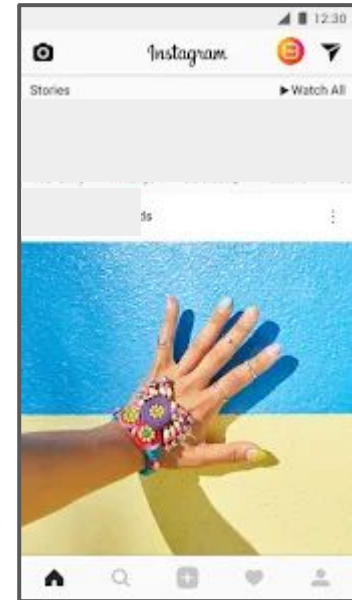
Content



videos



text



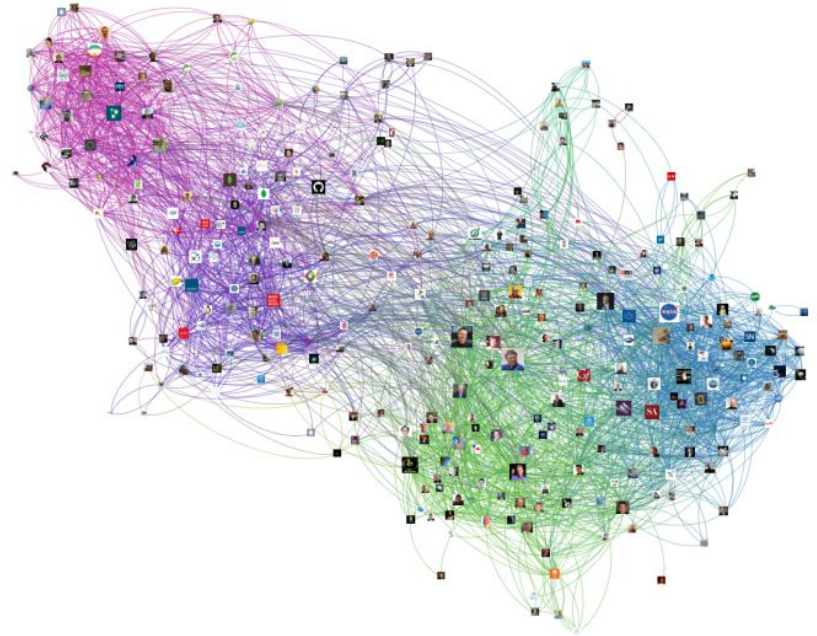
images

Prototypical Pipeline: Data Collection

collect data potentially containing tangible signals regarding our construct

Other data:

- **Connections** (friends, followers, links/URLs)
- **Actions** (likes, favs, comments)
- **Aggregate data** (views, shares, top searches)
- **User profiles** (with metadata)



Prototypical Pipeline: Data Collection

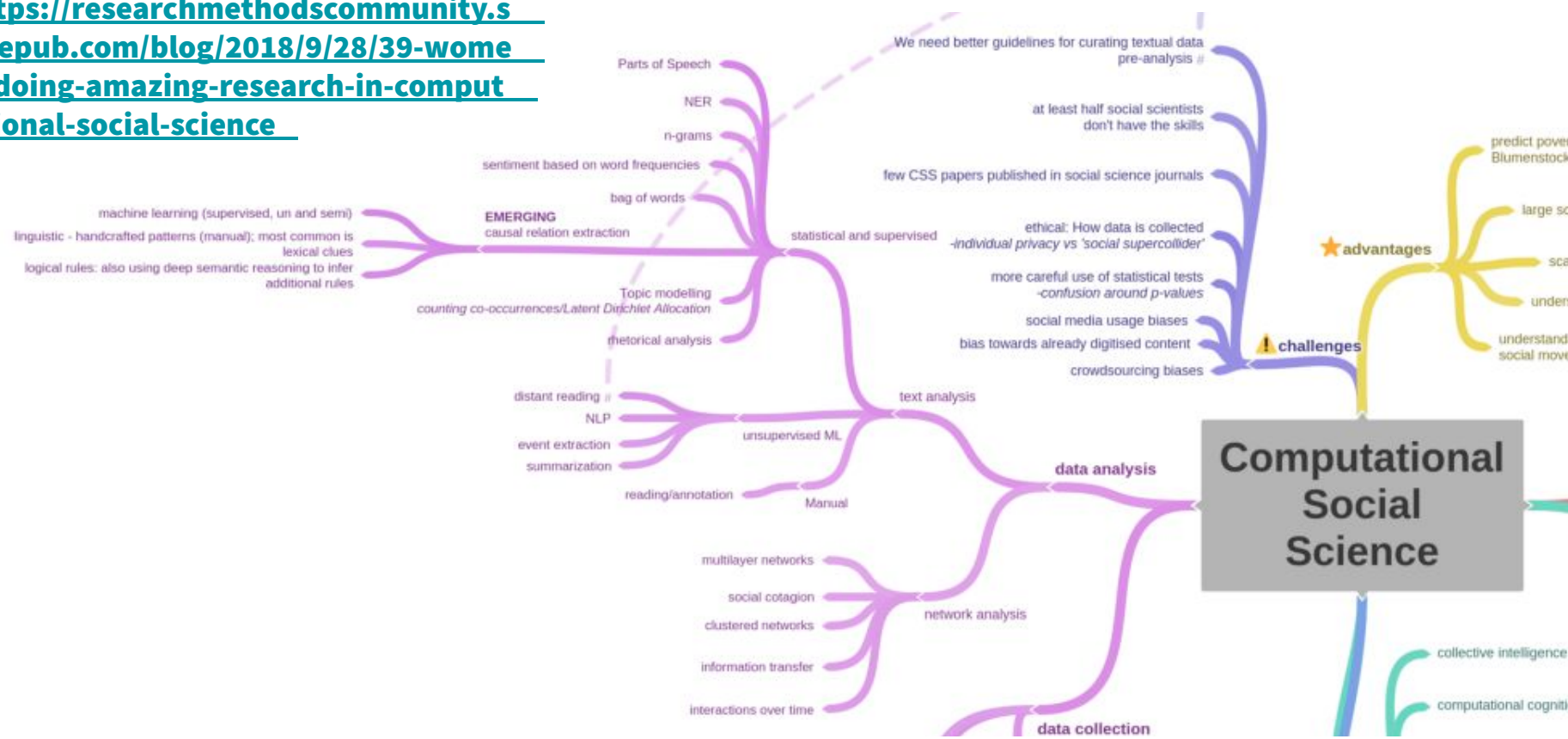
Source types:

- Application Programming Interfaces (API)
- Web scraping
- Web tracking
- Data donations
- Reuse existing datasets (data archives)
- Official sale
- Use 3rd party services (e.g. Pushsift, Crimson Hexagon)

What about data analysis?

A non-exhaustive mind map of computational social science, by Daniela Duca:

<https://researchmethodscommunity.sagepub.com/blog/2018/9/28/39-women-doing-amazing-research-in-computational-social-science>



A rough list of (inter-connected) methods

- Network analysis
- Content analysis
 - Text analysis
 - Image and video analysis
- Human Computer Interaction ('social computing')
- Causal inference
- Combinations of two or more approaches

Let's look at some concrete examples

Studying Polarization with Twitter

- **Construct:** political polarization in the U.S during mass shootings
- **Platform:** Twitter
- **Data collected:** 4.4 Million tweets with keywords related to mass shootings collected with the Twitter API
- **Analysis:** many methods of content analysis including topic modeling
- **Result:** Democrats and Republicans focus on different aspects during mass shootings

Demszky et al., [Analyzing Polarization in Social Media: Method and Application to Tweets on 21 Mass Shootings](#)

Analyzing Polarization in Social Media: Method and Application to Tweets on 21 Mass Shootings

Dorottya Demszky¹ Nikhil Garg¹ Rob Voigt¹ James Zou¹

Matthew Gentzkow¹ Jesse Shapiro² Dan Jurafsky¹

¹Stanford University ²Brown University

jurafsky}@stanford.edu

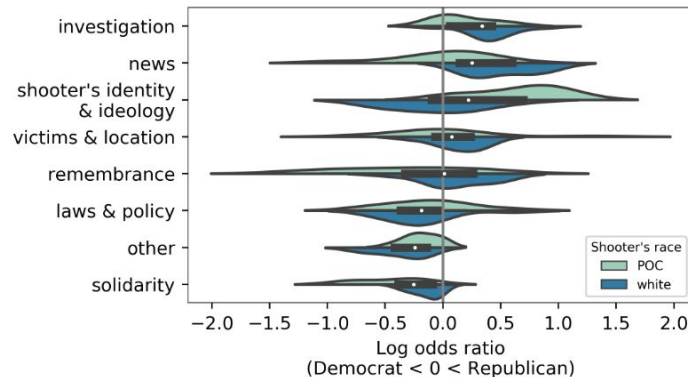


Figure 7: The plot shows the kernel density of the partisan log odds ratios of each topic (one observation per event). The white points show the median and the black rectangles the interquartile range across events.

(Bakshy et al., 2015). Prior to this, e.g., that polarized messages to be shared (Zafar et al., 2012); however, we lack understanding of the many ways that topics are more polarizing linguistically. more comprehensive framing of polarization by looking at topic choice, locutionary force.

Subjects of polarization by study-

Studying Politics with Tiktok vs. Youtube

- **Construct:** Political content
- **Platform:** TikTok and Youtube
- **Data collected:** 11K TikTok accounts who primarily post about politics and 1,712 political YouTube channels
- **Analysis:** descriptive summary of views, comments, and playbacks
- **Result:** Videos reach beyond the network more easily on Tiktok

Fifteen Seconds of Fame: TikTok and the Supply Side of Social Video

Benjamin Guinaudeau
University of Konstanz

Kevin Munger
Pennsylvania State University

Fabio Votta
University of Amsterdam

Abstract

TikTok has rapidly developed from a punchline for jokes about “kids these days” into a formidable force in American politics. The speed of this development is unprecedented, even in the rapidly-changing world of digital politics. Through a combination of hashtag and snowball sampling, we identify 11,546 TikTok accounts who primarily post about politics, allowing us to analyze trends in the posting, viewing and commenting behavior on 1,998,642 tiktoks they have uploaded. We test a number of theories about how the unique combination of affordances on TikTok shapes how it is used for political communication. Compared to the dominant platform for political videos (YouTube) we find that a higher

Guinaudeau, Benjamin, Kevin Munger, and Fabio Votta.

["Fifteen seconds of fame: TikTok and the supply side of social video."](#)

The End of Theory: The Data Deluge Makes the **Scientific Method** Obsolete

By Chris Anderson 06.23.08

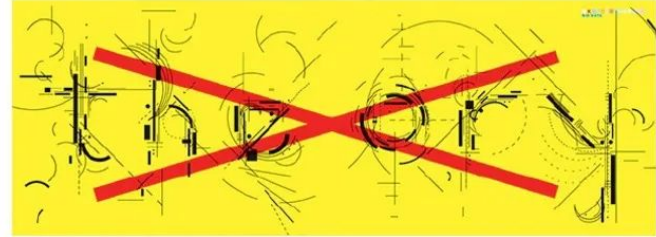


Illustration: Marian Bantjes

"All models are wrong, but some are useful."

So proclaimed statistician George Box 30 years ago, and he was right. But what choice did we have? Only models, from cosmological equations to theories of human behavior, seemed to be able to consistently, if imperfectly, explain the world around us. Until now. Today companies like Google, which have grown up in an era of massively abundant data, don't have to settle for wrong models. Indeed, they don't have to settle for models at all.

Sixty years ago, digital computers made information readable. Twenty years ago, the Internet made it reachable. Ten years ago, the first search engine crawlers made it a single database. Now Google and like-minded companies are sifting through the most measured age in history, treating this massive corpus as a laboratory of the human condition. They are the children of the Petabyte Age.

The Petabyte Age is different because more is different. Kilobytes were stored on floppy disks. Megabytes were stored on hard disks. Terabytes were stored in disk arrays. Petabytes are stored in the cloud. As we moved along that progression, we went from the folder analogy to the file cabinet analogy to the library analogy to — well, at petabytes we ran out of organizational analogies.

At the petabyte scale, information is not a matter of simple three- and four-dimensional taxonomy and order but of dimensionally agnostic statistics. It calls for an entirely different <https://www.wired.com/2008/06/pb-theory/> ^{e visualized in t for it later.}

For instance, Google conquered the advertising world with nothing more than applied mathematics. It didn't pretend to know anything about the culture and conventions of

Studying Social Media: No Big Deal?

Use case: Detecting the flu with digital traces

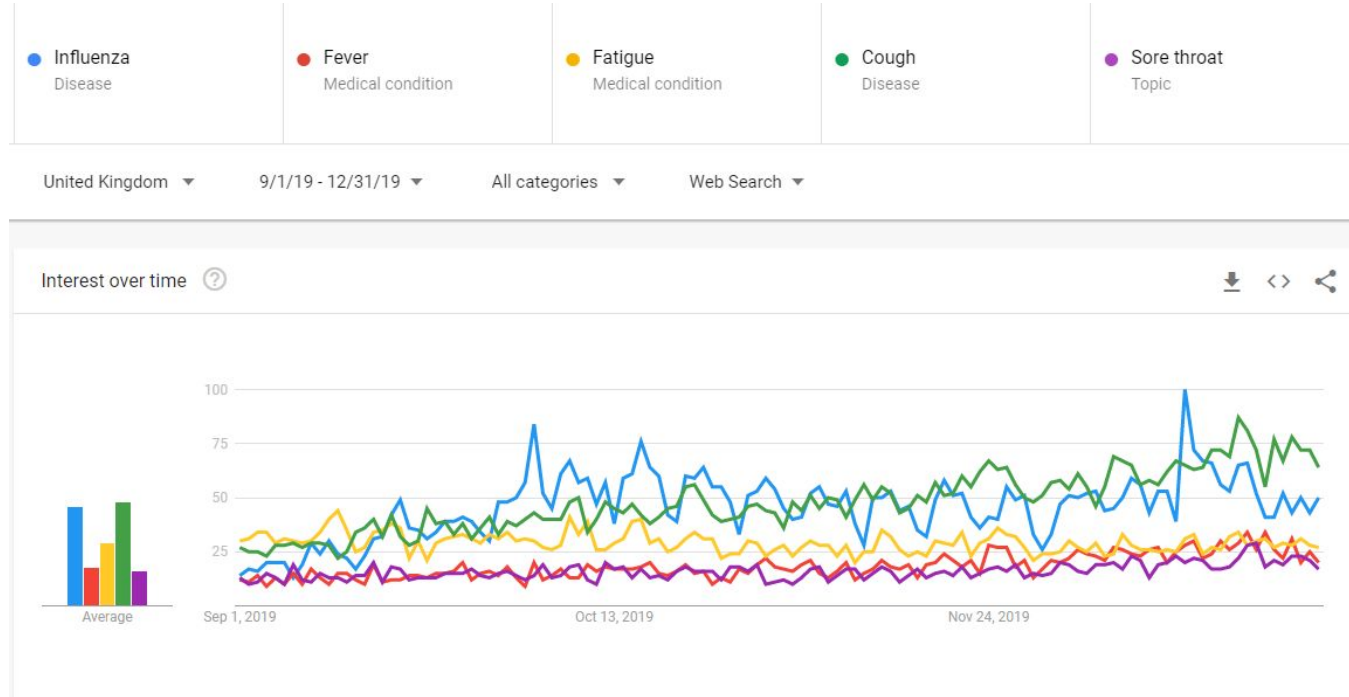
Google Flu (trends)



What proportion of US-Americans have the flu?

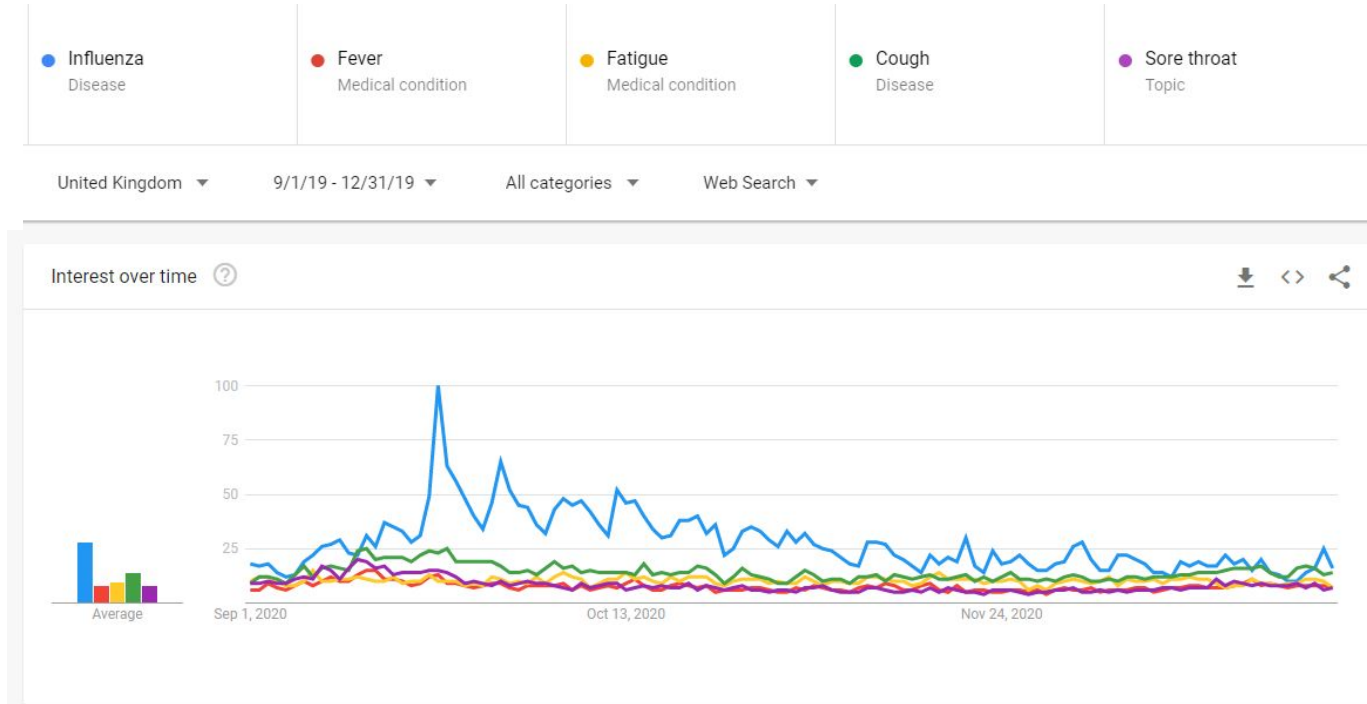
Approach

- Trends in Google searches related to influenza-like illnesses



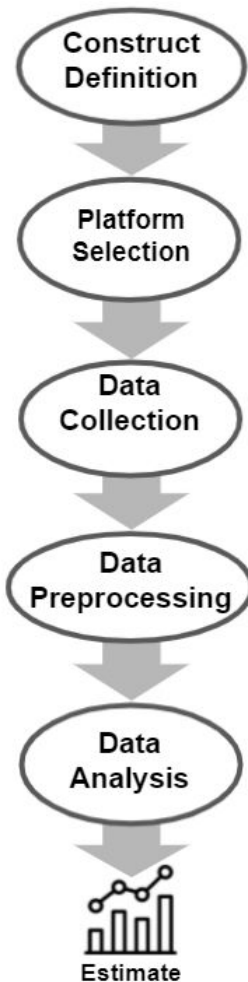
Approach

- Trends in Google searches related to influenza-like illnesses



Approach

- Construct: influenza-like illness (ILI)
- Target population(s): national and regional populations of the US
- Platform: Google web search
- Data collection: Logs 2003-2008, weekly, From 50 M. search terms, select those that best predict.
- Preprocessing: aggregated per each region, normalized by overall search activity, location via IP.
- Analysis: Regression where DV= ILI doctor visits, IV= ILI-related search fractions, for 1152 data points



Approach

□ Result

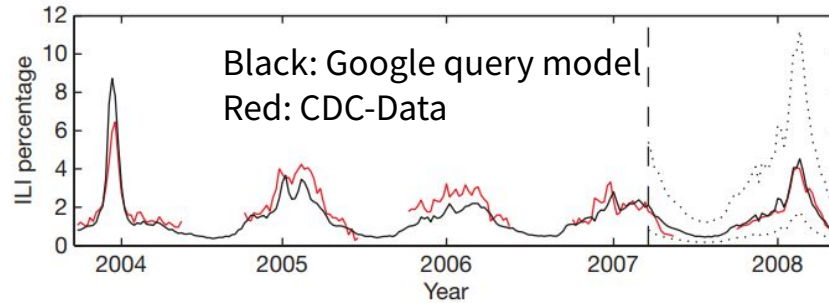
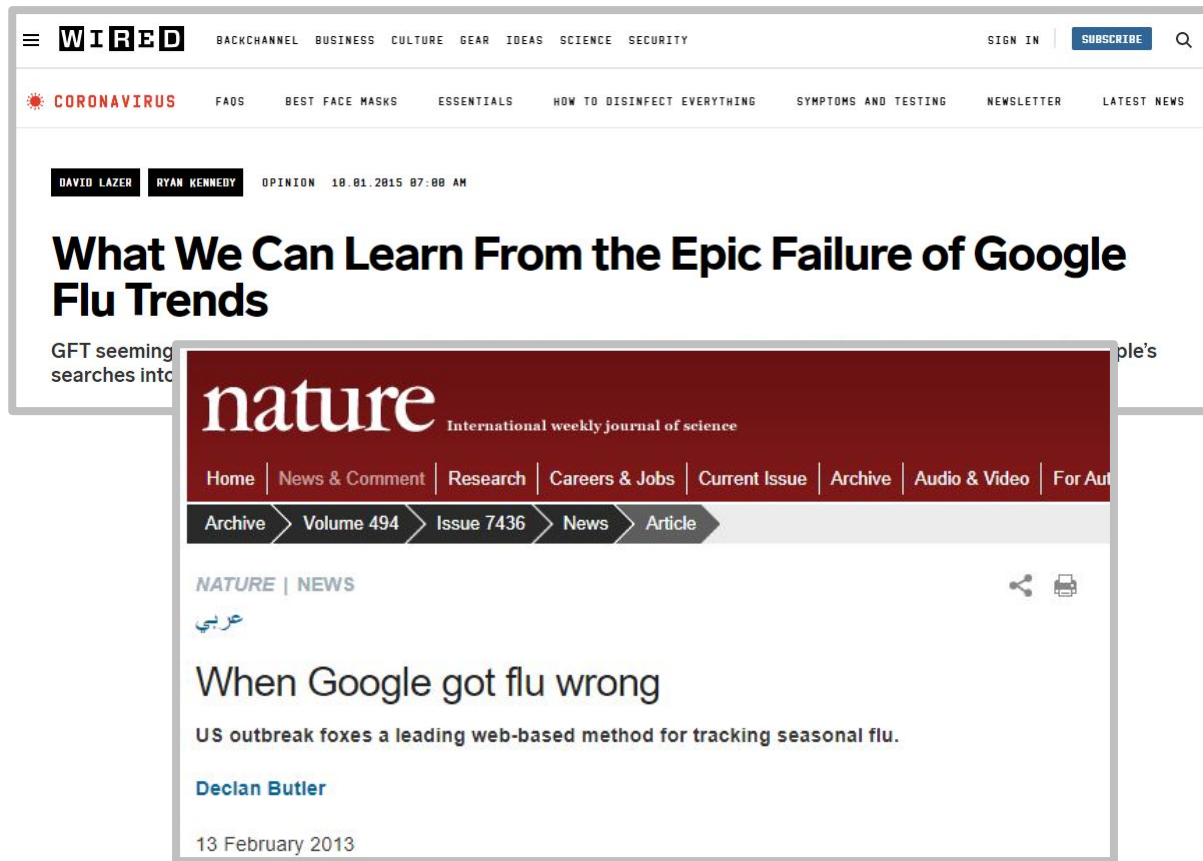


Figure 2 | A comparison of model estimates for the mid-2008 flu season (black) against CDC-reported ILI percentage (red). The model was fit over 128 points from 2004 to 2007, with a correlation of 0.96. The dotted line indicates 95% prediction intervals for New York, New Jersey and Pennsylvania.

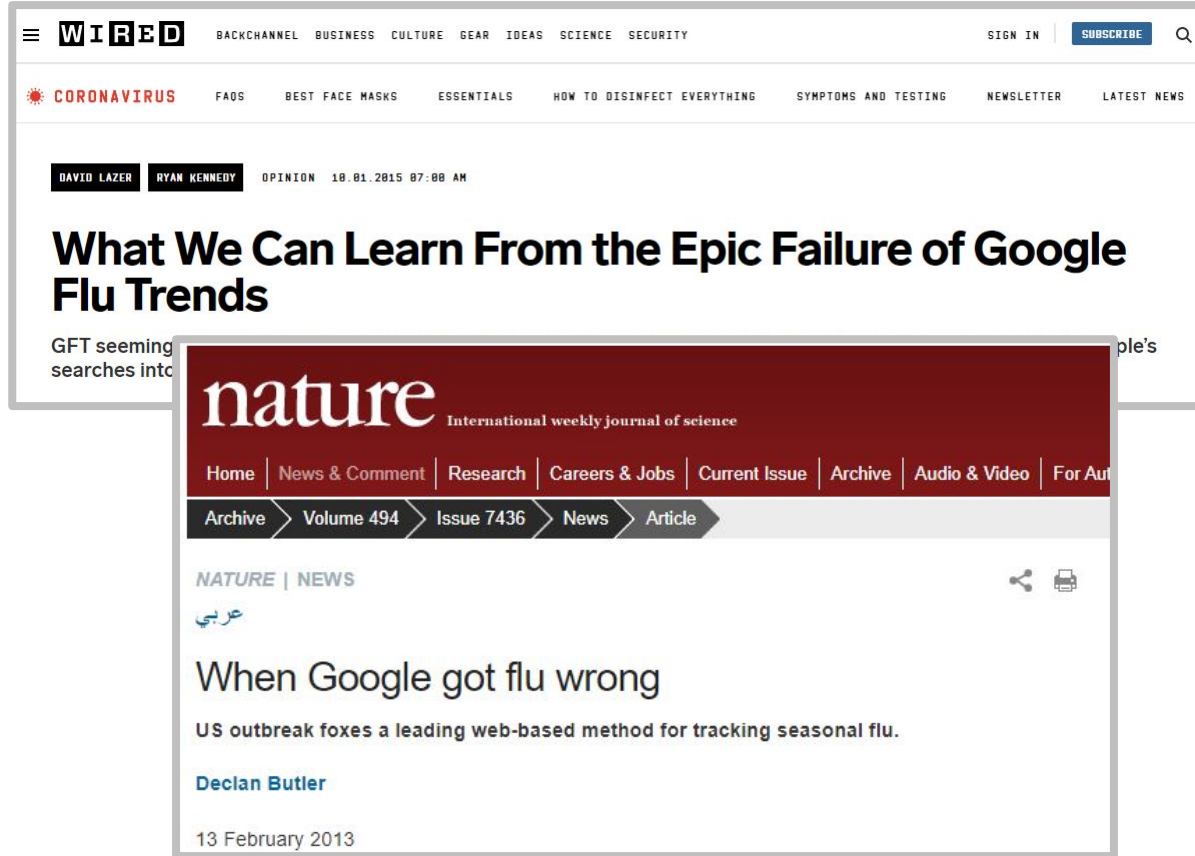
Problem solved

- Now possible: Now- or fore-casting, fine-grained local detection, other countries

The fail

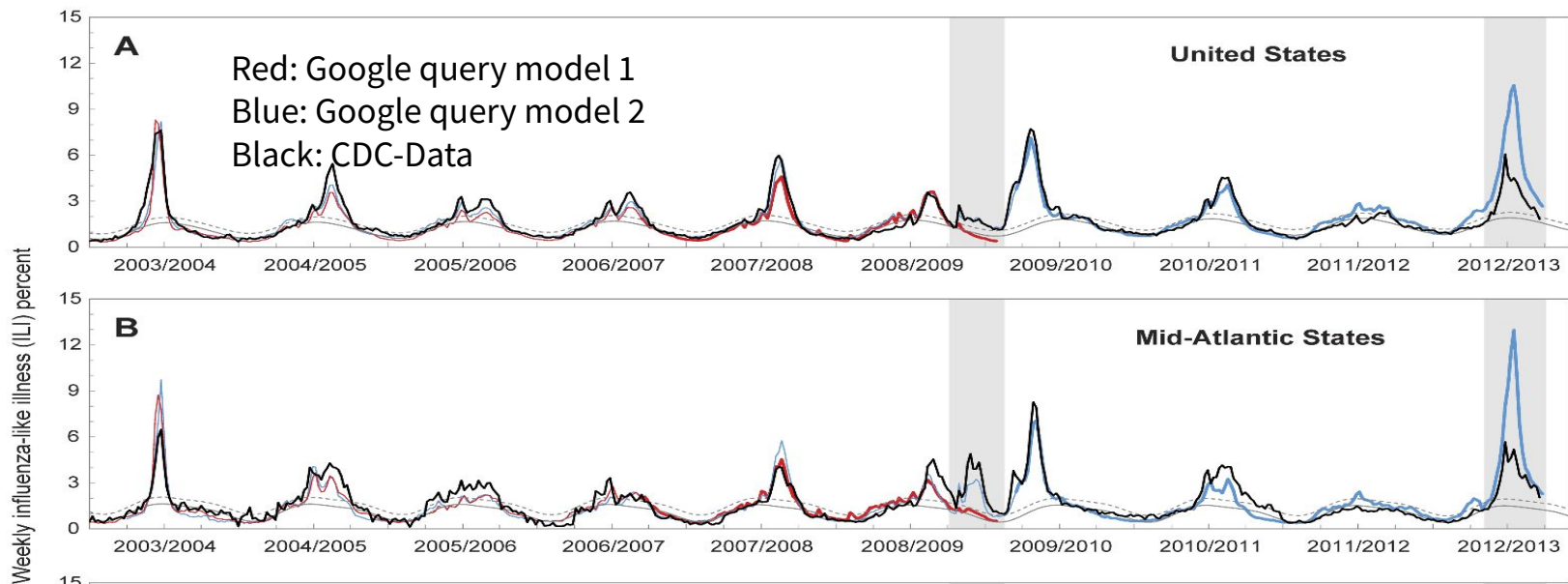


The fail



“In February 2013, Google Flu Trends (GFT) made headlines but not for a reason that Google executives or the creators of the flu tracking system would have hoped. Nature reported that GFT was predicting more than double the proportion of doctor visits for influenza-like illness (ILI) than the Centers for Disease Control and Prevention (CDC), which bases its estimates on surveillance reports from laboratories across the United States (1, 2).”

Overestimation



“part flue detector, part winter detector” □ errors are auto-correlated & direction and magnitude change with seasons

Where to be careful

“[...] there are enormous scientific possibilities in big data [but] the core challenge is that most big data that have received popular attention are not the output of instruments designed to produce valid and reliable data amenable for scientific analysis.”

Lazer, David, et al. "[The parable of Google Flu: traps in big data analysis.](#)" *Science* 343.6176 (2014)

Where to be careful

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“**Big data hubris** is the often implicit assumption that big data are a substitute for, rather than a supplement to, traditional data collection and analysis”

Lazer, David, et al. "[The parable of Google Flu: traps in big data analysis.](#)" *Science* 343.6176 (2014)

Social Media Data *complements* Traditional Data Sources

Article

The Price of Power: Power Seeking and Backlash Against Female Politicians

Tyler G. Okimoto¹ and Victoria L. Brescoll¹

Abstract

Two experimental studies were hypothesized to show that women may violate political norms, leading to backlash against their voting preferences. In Study 1, women's preferences were explained by their competence and the violation of community norms. In Study 2, the source of gender

Keywords

gender stereotypes, power, intention, moral outrage

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DOI: 10.1177/0146167210371940
http://pspb.sagepub.com

Full & Short Papers

Psychological Science, June 30–July 3, 2019, Boston, MA, USA

Bias in Digital Communication with Politicians

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1 INTRODUCTION

"Why do you even care for the #BER [Berlin airport] opening? Because you have to choose your clothes for the opening ceremony???"

This tweet directed at a female German politician resembles an often observed and reported phenomenon in our digital society: digital communication is driven by gender stereotypes rather than job-related content. This, in turn, can lead to bias and discrimination towards female professionals. Building on insights from research on social identities [60, 61] as well as gender biases and gender roles in

ABSTRACT

This study investigates gender bias in political interactions on digital platforms by considering how politicians present themselves on Twitter and how they are approached by others. Incorporating social identity theory, we use dictionary analyses to detect biases in individual tweets connected to the German federal elections in 2017. Besides sentiment analysis, we introduce a new measure of personal- vs. job-related content in text data, that is validated with structural topic models. Our results indicate that politicians' communication on Twitter is driven by party identity rather than gender. However, we find systematic gender differences in tweets

Social Media Data Analysis not necessarily superior to traditional data analysis.
But it's yet another methodology that helps you study society

The Price of Power: Power Seeking and Backlash Against Female Politicians

- Uses a survey experiment where people are shown fictional politicians with different traits

As the Tweet, so the Reply? Gender Bias in Digital Communication with Politicians

- Uses tweets mentioning real-life politicians and computationally assesses stereotypes in those tweets

Social Media Data Analysis

The Good 🤗

- Social Media provide a nice **complementary** data source to traditional social science data ([Lazer et al. 2009](#), [2020](#))
- The platforms themselves are an object of study
- Large scale data => an opportunity for data-driven approaches, like natural language processing (NLP)

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The Bad 😞

- Disproportionate focus on particular platforms / sources ([Tufekci 2014](#))
- Many representative biases ([Olteanu et al., 2021](#))
- Ad-hoc measurement models (“what are we *really* measuring?”)
- Need for a measurement theory ([Jungherr 2018](#))

Social Media Data Analysis

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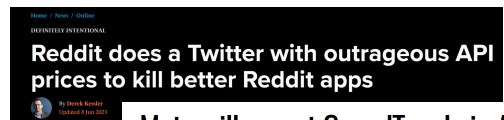
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The Ugly 🤢

- The API Apocalypse ([Freelon 2018](#); [Assenmacher et al., 2023](#))



Meta will sunset CrowdTangle in August

Advertisers and publishers will need to seek alternative platforms as CrowdTangle will no longer be available after August 14.

SCIENCE / TWITTER - X / TECH

Twitter just closed the book on academic research



/ Twitter was once an indispensable resource for academic research. That's changed under Elon Musk.



More on this next week

Beyond Methodological Questions: Ethics

- ❖ Research ethics practices are also still evolving
- ❖ Main focus on privacy
- ❖ Lack of informed consent
- ❖ “participants” not expecting research activities on their data
[[Fiesler & Proferes, 2018](#)]
- ❖ Different assumptions for different types of user groups (e.g. vulnerable groups)
- ❖ Potential starting point: [AoIR ethics guidelines](#)

Summary

- This session we learned:
 - About the computational turn in the Social Sciences
 - About Social Media and how they contribute to the Social Sciences
 - Prototypical research pipeline with Social Media Data
 - The Potentials and Pitfalls of research with Social Media Data

Summary and Next Week

- This session we learned:
 - About the computational turn in the Social Sciences
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 - Prototypical research pipeline with Social Media Data
 - The Potentials and Pitfalls of research with Social Media Data
- Next time:
 - What are the typical components of a social media platform?
 - Data Collection I: Web scraping social media data
 - Use Case: Collecting data from Truth Social
 - Readings:
 - Lazer, David MJ, et al. "[Computational social science: Obstacles and opportunities](#)," Science 369.6507 (2020): 1060-1062.
 - Gerard, Patrick, Nicholas Botzer, and Tim Weninger. "[Truth social dataset](#)," Proceedings of the international AAAI conference on web and social media. Vol. 17. 2023.

Additional Resources

- AwesomeCSS github page:
<https://github.com/gesiscss/awesome-computational-social-science?tab=readme-ov-file#online-courses-and-material>
- Bit by Bit: (you can read it online for free!):
<https://www.bitbybitbook.com/en/1st-ed/preface/>
- The Summer Institute in CSS: <https://sicss.io/>
- NLP + CSS tutorials:
<https://nlp-css-201-tutorials.github.io/nlp-css-201-tutorials/>

Questions?

We have a waitlist for the course,
so please deregister soon if you
know you don't want to take it.

Talk to me after the course if you had
problems registering